

INCIDENT REPORT

SSH Brute Force Attack — Ubuntu-Server-01

Homelab Security Operations | Kiara Earl

INCIDENT SUMMARY

Incident ID	IR-2026-001
Date / Time	May 24, 2026 — 00:55:32 UTC
Analyst	Kiara Earl
System Affected	Ubuntu-Server-01 (192.168.56.10)
Attack Type	SSH Brute Force / Credential Stuffing
Source IP	192.168.56.1 (Windows Host — simulated attacker)
Severity	MEDIUM
Status	RESOLVED — Attacker blocked by fail2ban
MITRE ATT&CK	T1110.001 — Brute Force: Password Guessing

PICERL INCIDENT RESPONSE

P — Preparation

Prior to the incident, the following controls were in place on Ubuntu-Server-01:

- SSH hardened — non-default port 2222, ed25519 key-based authentication, root login disabled, password authentication disabled (Exp002)
- UFW firewall active — only port 2222/tcp permitted inbound (Exp001)
- fail2ban deployed — configured to ban IPs after 3 failed SSH attempts within 5 minutes; 10-minute ban duration (Exp003)
- Splunk SIEM active — Universal Forwarder shipping auth.log and fail2ban.log to Splunk Enterprise on Windows host; SSH Attack Monitor dashboard and real-time ban alert configured (Exp004)

I — Identification

At 00:55:32 UTC, Splunk triggered a real-time alert: "fail2ban Ban Detected." Investigation of auth.log and fail2ban.log revealed the following sequence:

```
00:55:32 – Failed login attempt #1 from 192.168.56.1 (user: fakeuser)
00:55:33 – Failed login attempt #2 from 192.168.56.1 (user: fakeuser)
00:55:35 – Failed login attempt #3 from 192.168.56.1 (user: fakeuser)
00:55:36 – fail2ban NOTICE: Ban 192.168.56.1
```

Three failed password authentication attempts were made within 4 seconds from a single source IP. The attacker used a non-existent username (fakeuser) with forced password authentication — consistent with automated credential stuffing or brute force tooling. Password authentication had been disabled in Exp002; these attempts were blocked at the auth layer before any credentials were evaluated.

C — Containment

Containment was automatic. fail2ban inserted a UFW deny rule for 192.168.56.1 at 00:55:36 — 4 seconds after the first failed attempt. All subsequent connection attempts from that IP were dropped at

the firewall level.

- No manual intervention was required for initial containment
- Active SSH session from the banned IP was terminated immediately upon ban
- Splunk alert fired in real time, confirming SIEM detection pipeline is operational

E — Eradication

No malware, persistence mechanisms, or unauthorized access was established. The brute force attempts failed to authenticate — password authentication was disabled and the attacker did not possess a valid private key.

- No files were modified on the target system
- No unauthorized user accounts were created
- No lateral movement occurred — attacker never gained access to the system

R — Recovery

The ban expired after 10 minutes (600 seconds) per the configured bantime. Normal operations were not impacted — legitimate SSH access via port 2222 using key authentication remained available throughout the incident.

Ban manually removed for lab continuity:

```
sudo fail2ban-client set sshd unbanip 192.168.56.1
```

System integrity confirmed — UFW rules reviewed, no unauthorized rule changes detected, all services operating normally.

L — Lessons Learned

1. Defense in depth works — the layered controls (hardened SSH + UFW + fail2ban + Splunk) stopped the attack automatically with no manual intervention required.
2. Disabling password authentication is critical — even with brute force protection, eliminating password auth removes the attack vector entirely.
3. Real-time alerting is essential — the Splunk alert fired within seconds of the ban, enabling immediate analyst awareness without manual log review.
4. Banning the analyst's own IP terminates active sessions — in a production environment, ensure out-of-band access (console, IPMI) before testing ban rules.
5. Recommendation — consider reducing bantime to 30 minutes or implementing progressive banning (longer bans for repeat offenders) in a production environment.